

Macro Liquidity-Driven State-Switching Factor Rotation Strategy

Evidence from the Chinese A-Share Market

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1. Introduction

Motivation

Traditional **static** factor strategies fail in China's A-share market — policy interventions, retail-dominated investor structure, and monetary policy sensitivity cause **time-varying** factor performance.

Our Approach: Four Key Innovations

- ➊ **Macro Regime Classification** — 3-month Shibor → Risk-On / Risk-Off via median threshold
- ➋ **Dynamic Factor Weighting** — Rolling Spearman IC/IR by regime, IC threshold + momentum filter
- ➌ **Stratified Universe** — 120 stocks from CSI 300 (48/42/30 large/mid/small), top 25 monthly
- ➍ **Three-Layer Risk Management** — defensive allocation, volatility targeting, loss protection

2.1 Data & Universe Selection

Data (6 files, Feb 2015 – Dec 2024)

File	Content
macro_shibor	3-month Shibor (regime signal)
index_weights	CSI 300 constituent weights
stock_prices	OHLCV daily prices
daily_basic	PE, PB, dividend yield
fina_indicator	ROE, profit growth (quarterly)
index_prices	CSI 300 benchmark

IS: 2015–2019 | OOS: 2020–2024 | 2,411 trading days

Universe: Stratified Sampling

- 48 large + 42 mid + 30 small = **120 stocks**
- Top **25** selected monthly, equal-weighted
- In-sample rank-based — no look-ahead bias
- Larger cross-section → greater IC statistical power

2.2 Factor Definitions & Regime Classification

Six Factors

Factor	Calculation
Growth	$\frac{\text{Profit}_t - \text{Profit}_{t-4}}{ \text{Profit}_{t-4} }$
Value	$E/P_t + B/P_t$
Quality	$\frac{\text{Net Income}_t}{\text{Equity}_t}$
LowVol	$1/\sigma(r_{t-20:t})$
Dividend	$\frac{\text{DPS}_t}{\text{Price}_t}$
Momentum	$\frac{P_{t-20}}{P_{t-240}} - 1$

Winsorization (1%/99%) + z-score daily

Regime Classification

$$\text{Regime}_t = \begin{cases} \text{Risk-On} & \text{Shibor3M}_t < \text{Med}_{\text{IS}} \\ \text{Risk-Off} & \text{Shibor3M}_t \geq \text{Med}_{\text{IS}} \end{cases}$$

- Median from in-sample (2015–2019)
- **Risk-On:** 2,037 days (84.5%)
- **Risk-Off:** 374 days (Jan 2017 – Aug 2018)

2.3 Factor Weighting Scheme

Spearman Rank IC (robust to A-share outliers):

$$\rho_{\text{Spearman}} = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}, \quad \text{IC}_{f,t} = \rho_{\text{Spearman}}(F_{f,t}, R_{t+1})$$

Pooled on full 120-stock universe — no regime splitting — maximizes cross-sectional power
IC/IR Weight Formula (2-year rolling, by regime):

$$w_f = \max(0, \overline{\text{IC}}_f) \times \max\left(0, \frac{\overline{\text{IC}}_f}{\sigma(\text{IC}_f)}\right)$$

Composite Score:

$$\text{Score} = w_G F_G + w_V F_V + w_Q F_Q + w_L F_L + w_D F_D + w_M F_M$$

Safeguards: IC threshold = 0.005 | Momentum filter (> 20% ↓ → 50% cut) | Negative IC → zero | 40% cap

Default Weights (Fallback, IC history < 6 months):

Risk-On: G 0.05, V 0.15, Q 0.05, L 0.20, D 0.15, **M 0.40**

Risk-Off: G 0.00, V 0.20, Q 0.05, **L 0.35, D 0.20**, M 0.20

2.4 Backtesting & Risk Management

Backtesting Setup

Monthly rebalancing on last trading day, equal-weighted top 25.

Commissions 0.025% (one-way) | Stamp duty 0.10% (sell only) | Impact 0.05% (one-way)

→ **Round-trip: 25 bp**

Three-Layer Risk Management

- 1 **Defensive Allocation:** Equity $\rightarrow$ 50% when Shibor $>$ IS median; 50% $\rightarrow$ Shibor cash (grid-searched in-sample)
- 2 **Volatility Targeting:** Scale = $\min(1.5, \max(0.5, \sigma_{\text{target}}/\sigma_{t-60:t}))$, target $\sim 20\%$ ann.
- 3 **Loss Protection:** Monthly return $< -8\%$ $\rightarrow$ halve equity next month (circuit-breaker)

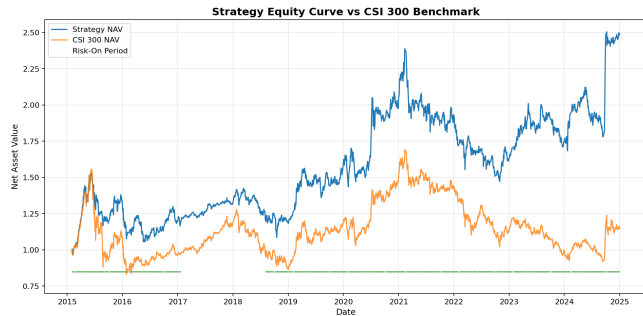
All overlays use contemporaneously observable information — zero look-ahead bias

3.1 Overall Performance

Metric	Full Sample	In-Sample	Out-of-Sample
Period	2015–2024	2015–2019	2020–2024
Total Return	148.53%	58.71%	56.59%
Annualized Return	9.98%	10.20%	9.77%
Annualized Volatility	20.06%	19.43%	20.67%
Sharpe Ratio	0.348	0.370	0.328
Sortino Ratio	0.571	0.581	0.561
Maximum Drawdown	-38.40%	-31.15%	-38.40%
Calmar Ratio	0.260	0.327	0.255
Information Ratio	0.707	0.554	0.842
Win Rate / Profit Factor	51.60% / 1.112	52.29% / 1.118	50.91% / 1.107

CSI 300 benchmark: 14.57% total, 1.43% annualized | **IS/OOS Sharpe near-parity → no overfitting**

3.2 Equity Curve & Drawdown

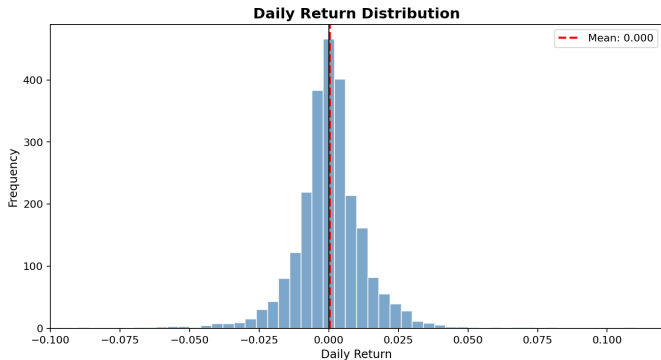


Max DD = -38.40% (2022 sell-off)

Strategy NAV = 2.485 vs. CSI 300 = 1.146

- Consistent outperformance across entire 10-year period; green dashed = Risk-On
- IS max DD -31.15% (2015 crash), OOS -38.40% (2022); three overlays work in concert

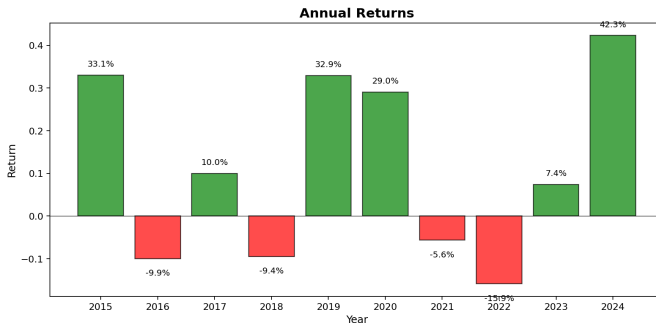
3.3 Return Distribution



Statistic	Value	Statistic	Value
Skewness	0.05 (near-zero)	VaR _{95%}	-1.85%
Kurtosis	7.31 (fat tails)	CVaR _{95%}	-2.95%

Near-zero skewness → risk overlays contain left-tail; elevated kurtosis from vol-targeting leverage

3.4 Annual Returns



2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
+34.95%	-4.79%	+8.96%	-10.39%	+34.16%	+26.12%	-6.49%	-14.74%	+6.84%	+42.02%

6 out of 10 years positive (60% win rate) | Worst: -14.74% (2022) | Best: +42.02% (2024)

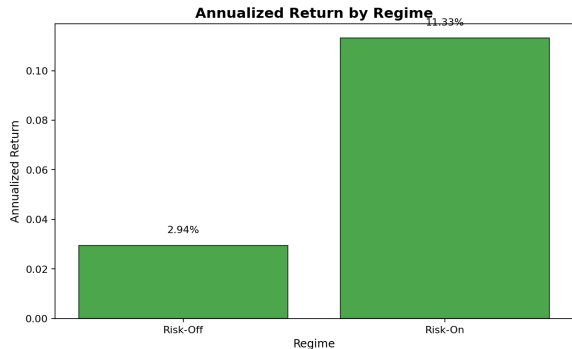
3.5 Rolling Sharpe Ratio



- 1-year rolling Sharpe (252-day), predominantly positive throughout
- Strong positive: 2015, 2019–2020, 2024; briefly negative: 2018, 2021
- OOS period (2020–2024): predominantly positive, confirming robustness

3.6 Regime Performance

Metric	Risk-Off	Risk-On
Period	2017/01–2018/08	2015/02–2024/12
Days	374	2,037
Total Return	+4.40%	+138.06%
Ann. Return	+2.94%	+11.33%
Ann. Volatility	9.98%	21.40%
Sharpe Ratio	-0.006	0.389
Max Drawdown	-13.80%	-38.40%
Final NAV	1.044	2.381



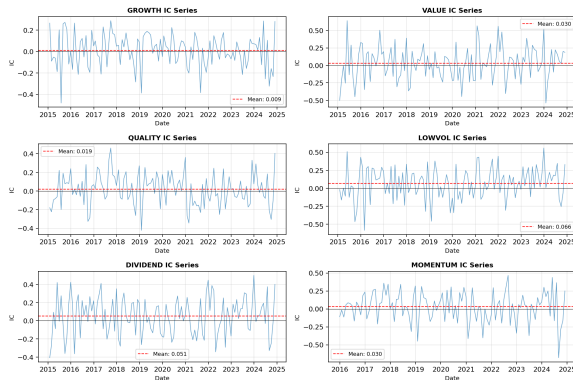
Risk-Off: 50% equity + 50% cash | Risk-On: full equity

3.6 Factor IC Analysis

IC Stability (Pooled, Full Sample)

Factor	Mean IC	IC Stab.	t-stat
LowVol	0.066	0.301	3.279***
Dividend	0.051	0.245	2.678***
Momentum	0.030	0.144	1.499
Value	0.030	0.126	1.374
Quality	0.019	0.110	1.197
Growth	0.009	0.061	0.665

*** 1% significance. Power from 120 stocks + Spearman IC.



LowVol: most consistently positive IC, especially during stress

3.7 Factor Weights Evolution



Risk-On evolution: Momentum-dominant (2015–16) → Balanced (2017–21) → Value + LowVol + Dividend (2022–24)

Risk-Off: mostly defaults (limited 374-day sample); filtered by IC threshold & momentum filter

4.1 Sensitivity Analysis

IC Window Length					
Window	1 year	2 years	3 years	4 years	5 years
Sharpe	0.273	0.348	0.304	0.085	0.052
Number of Stocks					
N	10	15	20	25	30
Sharpe	0.188	0.247	0.252	0.348	0.308
Transaction Costs					
Scenario	Low (16bp)	Base (25bp)	High (40bp)	V.High (70bp)	
Sharpe	0.365	0.348	0.321	0.266	

2-year IC window optimal (A-share IC relationships are time-varying) | **25 stocks** optimal | Sharpe > 0.26 even at 70bp

4.2 Additional Risk Metrics

Metric	Value	Metric	Value
Avg Monthly Turnover	28.24%	Strategy Beta	0.764
Up Capture Ratio	80.6%	Down Capture Ratio	73.4%
Skewness	0.05	Kurtosis	7.31
VaR _{95%}	-1.85%	CVaR _{95%}	-2.95%
Correlation with Market	0.839	Profit Factor	1.112

- **Asymmetric capture:** 80.6% upside vs 73.4% downside — risk overlays at work
- Beta $0.764 < 1.0$ — alpha not from levered market exposure

5. Conclusion

1 Strong Risk-Adjusted Performance

148.53% total, Sharpe 0.348 — CSI 300:
14.57%

2 Significant Factor IC

LowVol ($t=3.28^{***}$), Dividend ($t=2.68^{***}$) at
1%

3 No Overfitting

IS Sharpe 0.370 / OOS Sharpe 0.328

4 Risk Management Works

Beta 0.764, Down capture 73.4%

Limitations

- Single Risk-Off episode (374 days)
- 2/6 factors significant at 1%
- Single Shibor indicator
- Vol-targeting amplifies DD

Future Work

- Bayesian shrinkage for IC estimation
- Multi-dimensional regimes (credit, PMI, M2)
- ML for non-linear factor interactions

Thank You

Questions & Discussion

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